Intergenerational Housing Misallocation and Fertility

A lifecycle model of access to family-sized housing

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Family formation requires access to space

Children increase a household's demand for living space, but family-sized housing is unevenly allocated across generations.

- Large rental units are scarce.
- Buying a larger home requires liquid wealth and a down payment.
- Older households retain much of the family-sized owner stock after children leave.

Question: Does limited access to family-sized housing raise the cost of children and reduce family formation among young households?

Source: Circulated note, Introduction.

The paper connects access, allocation, and fertility

1. A two-generation analytical model isolates how rental limits, down payments, and old-owner retention affect the shadow value of space.
2. A finite-life quantitative model adds income risk, saving, tenure choice, transaction costs, bequests, entry, and upward-sloping housing supply.
3. Counterfactuals compare a purchase grant for family-sized homes with a package that combines the grant and a higher property tax.

The central object is the access gap: the amount by which a constrained household values another unit of family space above its market user cost.

Source: Circulated note, Introduction and status of the draft.

A fixed housing stock is allocated across generations

Young households

- Draw income and liquid wealth, $i = (y_i, b_i)$.
- Choose entry, tenure, housing, completed fertility, and saving.
- Become old owners or old renters next period.

Old households

- Owners choose how much housing to retain and the estate to leave.
- Renters demand housing without an ownership lock-in wedge.
- All households exit after old age.

One stock $\bar{H}$ clears at rental user cost $r = \rho P$. Renting is capped at $\bar{h}^R$ and owning at $\bar{h}^O$, with $\bar{h}^R < \bar{h}^O$.

Source: Circulated note, Analytical Model: Environment.

Children use both goods and housing

A young household receives utility from adult consumption, residual space, and completed fertility:

$$u_i^Y = \log(c_i - \chi n_i) + \alpha \log(h_i - \kappa n_i) + \vartheta \log n_i.$$

- Each child requires χ units of consumption and κ units of housing.
- Renting and owning have the same market user cost per unit of space.
- Tenure matters because only ownership reaches the largest units and buying requires equity.

A buyer must satisfy $(1-\phi)Ph_i \leq b_i$: current liquid wealth, not lifetime income, governs access to owner housing.

Source: Circulated note, Analytical Model: Households.

Rental limits and financing collapse into an effective cap

Using $P = r/\rho$, the maximum housing available in each tenure is

$$H_i^R = \bar{h}^R, \qquad H_i^O = \min\left\{\bar{h}^O, \frac{b_i \rho}{(1-\phi)r}\right\}.$$

- Renters are constrained by the largest available rental unit.
- Owners may be constrained by the physical menu or by the down payment.
- A household can afford the flow cost of space yet remain unable to purchase it.

The relevant friction is therefore not tenure by itself, but the household-specific cap on family-sized space.

Source: Circulated note, Analytical Model: Households and equilibrium policies.

A binding housing constraint raises the full price of a child

For a constrained household, the marginal value of space exceeds its effective price by $\zeta_i^m \geq 0$:

$$\frac{\alpha(c_i^m - \chi n_i^m)}{h_i^m - \kappa n_i^m} = r^m + \zeta_i^m.$$

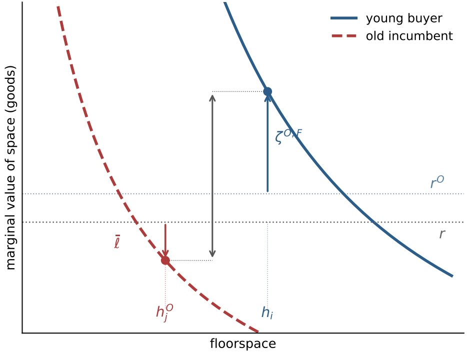
The fertility first-order condition becomes

$$\underbrace{\frac{\vartheta(c_i^m - \chi n_i^m)}{n_i^m}}_{\text{willingness to pay for a child}} = \underbrace{\chi + \kappa(r^m + \zeta_i^m)}_{\text{full price of a child}}.$$

Interpretation: limited access to space acts like a household-specific tax $\kappa\zeta_i^m$ on children, even when market prices do not change.

Source: Circulated note, Analytical Model: Equilibrium policies.

The competitive allocation can leave gains from reallocation



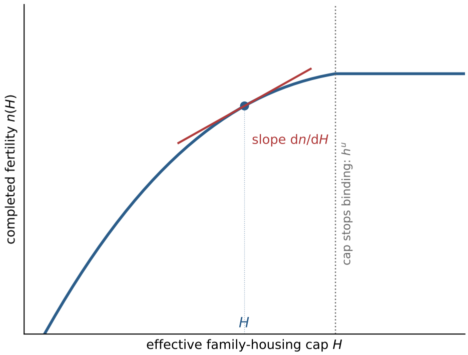
Moving one unit from a locked-in old incumbent to a financing-constrained young buyer produces surplus

$$\zeta_i^{O,F} + q\bar{\ell} + \bar{\ell} > 0.$$

- The buyer values space above effective user cost.
- The incumbent values retained space below the market rent because selling realizes a taxable gain.
- The common market price cancels.

Source: Circulated note, Proposition 1 and Figure 1.

Relaxing the cap can increase completed fertility



Source: Circulated note, Analytical Model: Fertility and policy; Figure 2.

Along a binding cap, more space has two effects:

- it makes the child's housing requirement easier to meet;
- it tightens the budget through the cost of additional space.

Fertility rises when the housing channel dominates:

$$\chi \leq (1+\Lambda)\frac{\kappa r^m}{\alpha}.$$

Exposure and pass-through determine the aggregate effect of a housing policy.

The quantitative model follows households over the lifecycle

18	18–42	66	85
entry	fertile window	retirement	maximum age

- Households face persistent earnings risk and save in liquid assets.
- A childless household makes a one-time discrete completed-family-size choice; children then age and leave home.
- Each period households choose renting or one of five owner sizes and adjust saving.
- Entry and the stationary population scale respond in counterfactuals.

The model preserves the analytical mechanism while adding the lifecycle margins needed for measurement and policy.

Source: Circulated note, Quantitative Lifecycle Model: Lifecycle and income.

Family needs enter a Stone–Geary composite

Dependent children shift minimum consumption and housing requirements:

$$\bar{c}(n,s) = \bar{c}_0 + \bar{c}_n n_s, \qquad \bar{h}(n,s) = \bar{h}_0 + \bar{h}_{\text{jump}} \mathbf{1}\{n_s \geq 1\} + \bar{h}_n n_s.$$

Flow utility is CRRA over residual consumption and usable housing services:

$$u = \frac{\left[(c-\bar{c})^{\alpha_c} \mathbf{h}^{1-\alpha_c}\right]^{1-\sigma}}{1-\sigma} + \psi_{\text{child}} n_s,$$

where owners receive service premium χ_O after household needs are removed.
At death, households value liquid wealth plus the gross value of the house through a warm-glow bequest motive.

Source: Circulated note, Quantitative Model: Preferences and bequests.

Ownership expands the housing menu but requires liquidity

Renters

- Choose rooms continuously up to $\bar{h}^R = 6$.
- Cannot borrow: $b' \geq 0$.
- Pay rental user cost r.

Owners

- Choose $h \in \{2, 4, 6, 8, 10\}$ rooms.
- Can borrow up to $\phi = 0.80$ of house value.
- Pay depreciation, property tax, and a 6% sale cost when adjusting.

Housing supply is upward sloping,

$$H^S(r) = \bar{H}\left(\frac{r}{\bar{r}}\right)^{\eta}, \qquad \eta = 1.75,$$

so higher demand is absorbed by both quantities and prices.

Source: Circulated note, Quantitative Model: Housing, finance, and taxes.

Family size, tenure, and housing are chosen jointly

A childless household in the fertile window chooses family size first, then its current housing position:

$$n' \in \{0, 1, 2\}, \qquad h' \in \{0, 2, 4, 6, 8, 10\}.$$

- $h' = 0$ means renting; positive values are owner rungs.
- A positive family-size choice fixes completed fertility and activates dependent-child needs immediately.
- Type-I extreme-value shocks smooth both family-size and tenure choices.
- The equilibrium combines optimal policies, the stationary distribution, entry and population scale, and housing-market clearing.

A birth can therefore change tenure and occupied space in the same period, rather than only affecting future housing demand.

Source: Circulated note, Quantitative Model: Household problem and stationary equilibrium.

Calibration combines external restrictions with 15 moments

Assigned outside estimation

- Lifecycle timing and survival
- $\sigma = 2$, real return, taxes, depreciation
- $\phi = 0.80$ and sale cost 6%
- Income process and housing menus
- Supply elasticity and entrant wealth

Estimated internally

- Patience and consumption weight
- Consumption and housing needs
- Value and dispersion of family size
- Ownership premium and tenure dispersion
- Housing-supply scale and bequests

Fourteen parameters are jointly disciplined by CPS fertility, ACS housing and tenure, and PSID wealth and estate moments.

Source: Circulated note, Calibration: Tables 1 and 2.

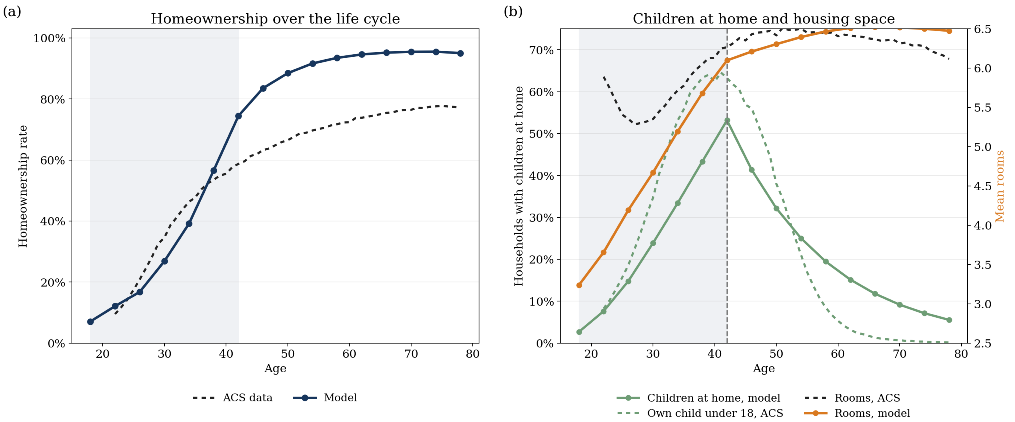
The fit reveals a lifecycle tenure tension

Moment	Data	Model
Completed fertility / childlessness	1.918 / 0.188	2.034 / 0.189
First-child rooms increment	0.664	0.681
Family ownership gap	0.168	0.219
Ownership, ages 30–55	0.575	0.658
Ownership, ages 25–34	0.341	0.274
Ownership, ages 65–75	0.764	0.954
Renter rooms / owner–renter rooms gap	3.805 / 2.419	3.963 / 2.323
Childless-owner share with ≥ 6 rooms	0.596	0.760
Young liquid wealth / income	0.179	0.328
Estate median / income	6.501	6.431
Older nonhousing-wealth share	0.608	0.610

The largest tension is excessive persistence in homeownership: young households buy too late, while older households rarely exit ownership.

Source: Circulated note, Calibration: Target moments and fit.

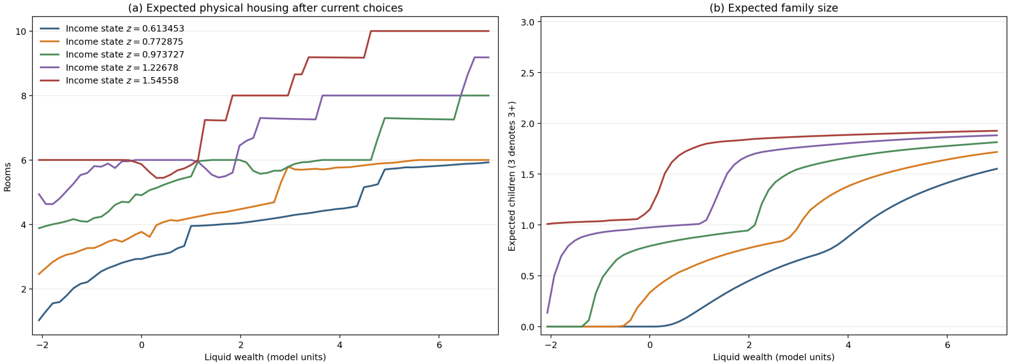
Lifecycle housing and child profiles are shifted too late



Homeownership is understated through the mid-thirties and overstated from the early forties onward. The child-at-home profile has the right hump but is delayed and too persistent.

Source: Circulated note, Lifecycle Housing Patterns, Figure 3.

Wealth and income jointly govern family size and housing



At age 42, higher-income households choose children and larger homes at lower liquid-wealth levels. The discrete owner ladder generates sharp changes in physical rooms.

Source: Circulated note, Lifecycle Housing Patterns, Figure 4.

Constrained renters value another room far above user cost

Among fertile renters whose positive-family branch reaches the rental cap:

$$\underbrace{r}_{0.130} + \underbrace{\zeta_i^R}_{0.178} = \underbrace{0.307}_{\text{marginal value of another room}} = 2.37\, r.$$

Holder of owner homes with ≥ 6 rooms	ACS	Model
Households age 46+	71.8%	80.7%
Households with no children at home	54.9%	74.8%

The households facing the highest marginal value of family space are distinct from those holding most of the family-sized owner stock.

Source: Circulated note, Quantitative Evidence on the Mechanism.

Policies that relax purchase access raise total births

Policy	Δ total births	Δ housing price
Purchase grant for family-sized homes	**+2.1%**	+0.90%
Higher property tax plus purchase grant	**+2.7%**	-18.44%

- The grant covers roughly one-third of the required down payment on an eligible home.
- The package also raises the annual property-tax rate on owner housing from 1 to 2 percent.
- Entry, population, and the housing price adjust in both exercises.

Source: Circulated note, Housing Policy, Table 5.

Housing affects fertility when policy changes access to space

- Children make family-sized housing valuable precisely when young households are least able to access it.
- Rental-size limits and down payments create an access gap that raises the effective price of children.
- Older households hold most large owner homes; with retention wedges, a cleared market need not allocate those homes efficiently.
- Purchase assistance raises births in the quantitative model, especially when paired with a policy that changes incumbent housing demand.

Current limits: fertility is one-shot rather than sequential, income risk is provisional, old-age ownership is too persistent, and the one-market model cannot separate new births from births relocated across markets.

Source: Circulated note, Conclusion and limitations.